Task 3: Modelling Budgets and Downside Risk

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1 Task 3: Modelling Budgets and Downside Risk

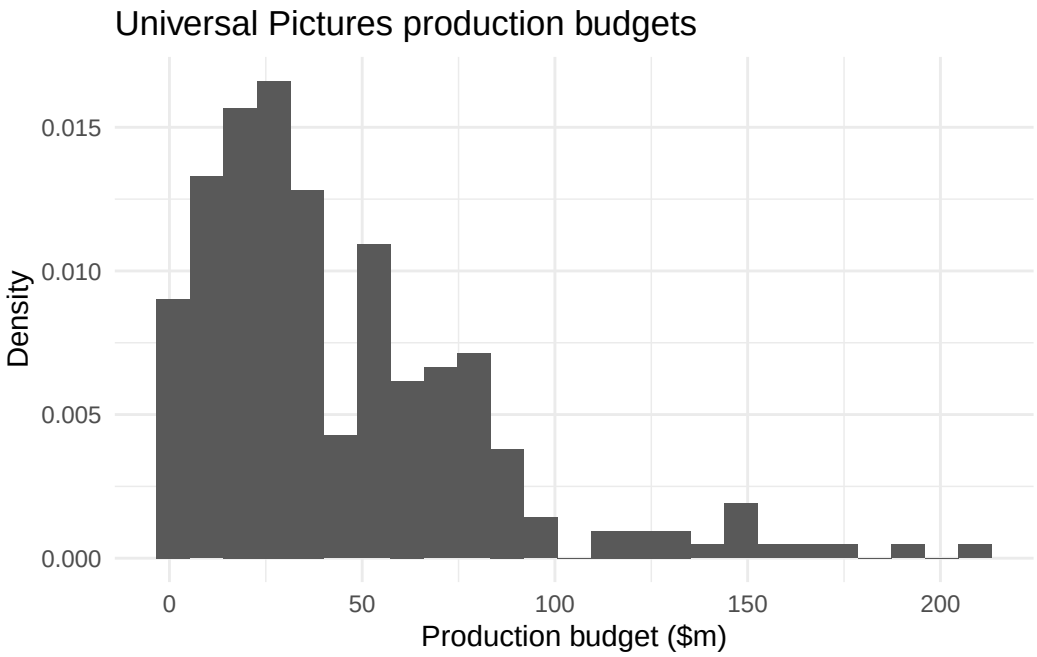
1.1 Introduction

[Explain the two financial planning questions.]

1.2 Universal Pictures Budget Modelling

1.3 Universal Pictures Budget Modelling

[Document the distributions considered.]



1.4 Model Diagnostics and Comparison

Distribution	Parameters (units)	Estimate 1	Estimate 2	Log-likelihood	AIC
Gamma	shape, rate (per $m)	1.382	0.031	-1,158.1	2,320.1
Weibull	shape, scale ($m)	1.214	47.479	-1,158.3	2,320.6
Log-normal	meanlog, sdlog (log $m)	3.392	1.015	-1,172.6	2,349.2
Normal	mean, sd ($m)	44.481	37.432	-1,225.1	2,454.1

Table 1: Distributions fitted to Universal Pictures budgets, ordered by AIC (lowest first)

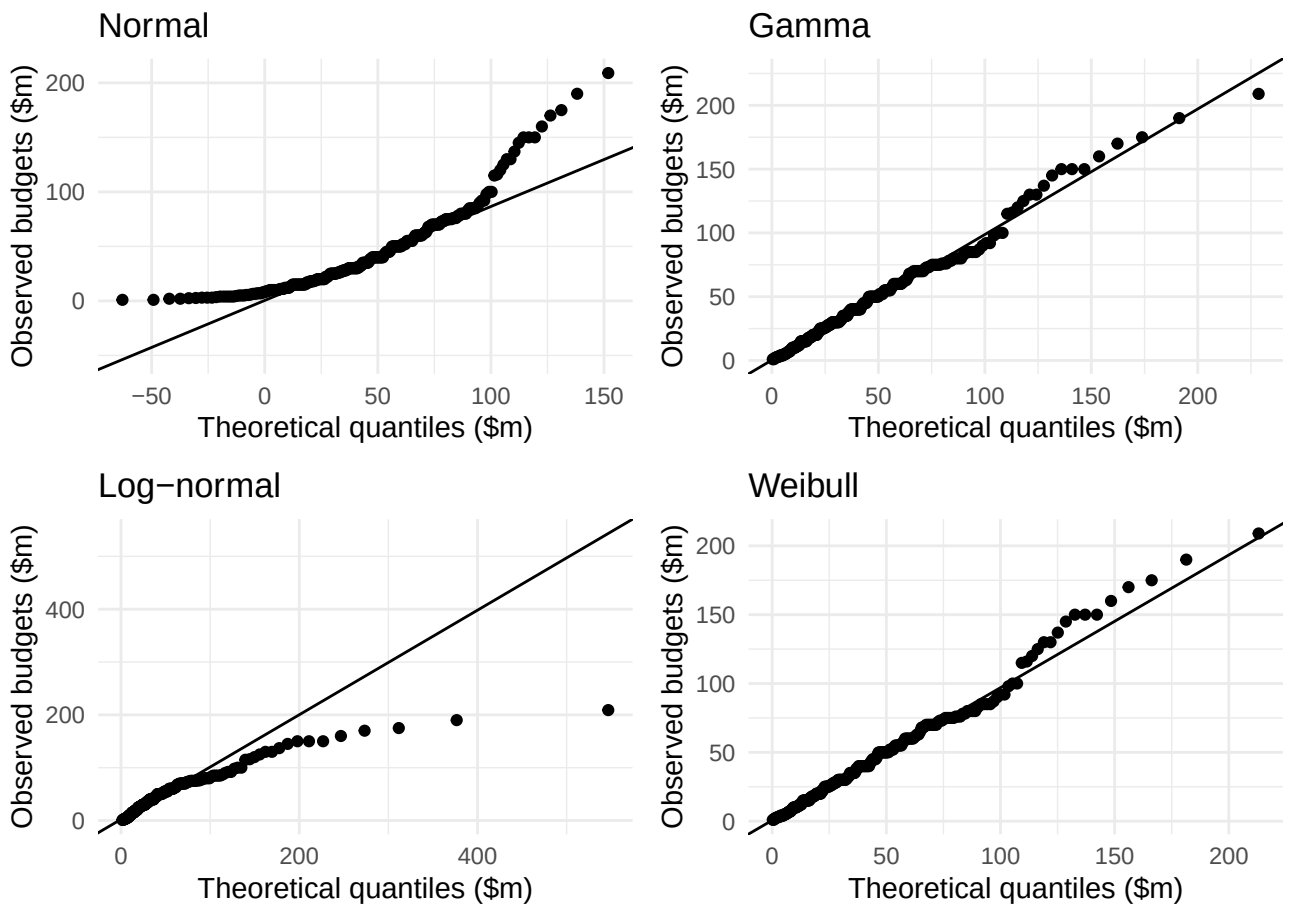


Figure 1: Q-Q plots of Universal Pictures budgets against each fitted distribution. Points close to the line indicate a good fit.

1.5 Action Genre Downside Risk

Method	25th percentile of ROI	95% lower	95% upper
Percentile bootstrap	0.91	0.80	1.05
Fitted log-normal model	0.73	0.66	0.82

Table 2: 95% interval estimates for the 25th percentile of ROI for Action films, from two methods

1.6 Comparison of Interval Estimates

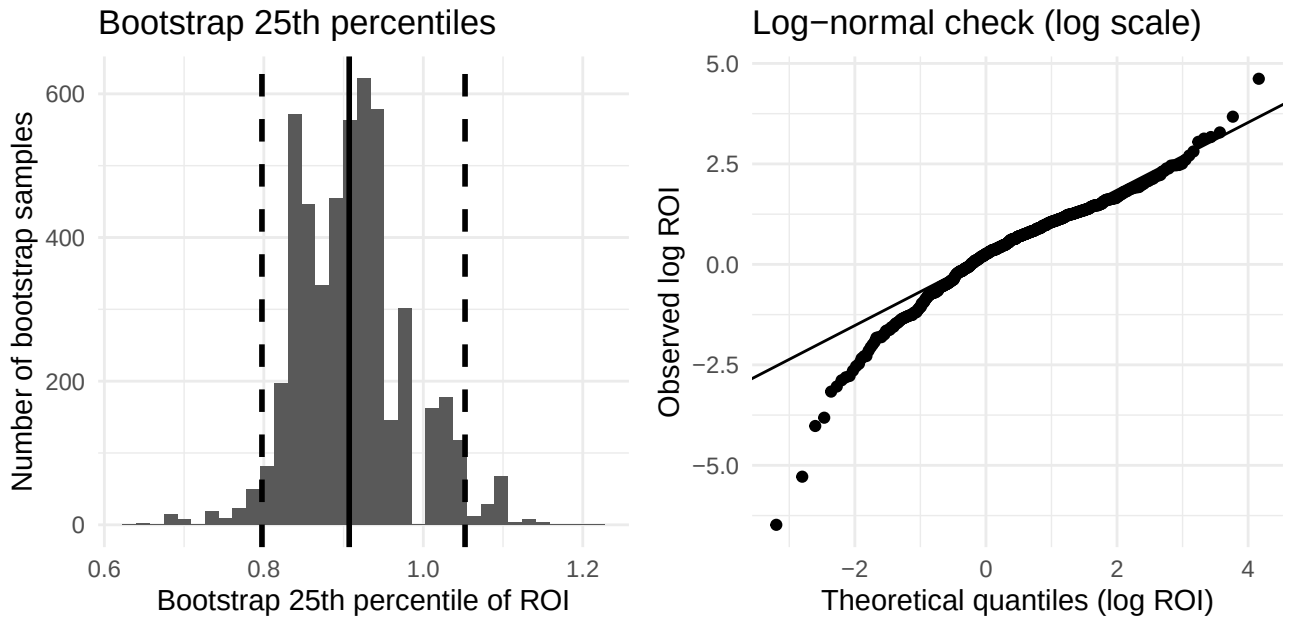


Figure 2: Left: bootstrap distribution of the 25th percentile of Action-film ROI (solid line: sample estimate; dashed lines: 95% interval). Right: Q-Q plot of log ROI against the fitted log-normal model.

1.7 Summary for the Finance Team

[Summarise the implications in plain language.]